

## BCT-SMOKE

**BCT-SMOKE: OHC Resonance-Guided Selective Pyrolysis**

*Geometric Suppression of Polycyclic Aromatic Hydrocarbon Formation  
via D4 Lattice Frequency Matching and Void-Geometry Filter Architecture*

*“The story so far: in the beginning, the universe was created. This has made a lot of people very angry and been widely regarded as a bad move. Nobody, however, had yet thought to ask the universe why it insisted on producing carcinogens when perfectly good geometry was available.”*

— Douglas Adams (slightly paraphrased by the geometry)

Michel Robert Cabrié

Independent Artist & Researcher, Barrys Reef, Victoria, Australia

ORCID: 0009-0007-9561-9859 · ZeroFreeParameters@gmail.com

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Tobacco smoke contains over 70 confirmed carcinogens, principally polycyclic aromatic hydrocarbons (PAHs) formed during incomplete combustion, and tobacco-specific nitrosamines (TSNAs) formed from alkaloid precursors. Both compound classes arise from specific pyrolytic reaction pathways governed by molecular bond geometry. The BCT Superfluid Lattice Model identifies two geometric intervention mechanisms: (i) OHC resonance-guided selective pyrolysis — application of D4 lattice-matched electromagnetic frequencies during combustion to disrupt PAH ring-closure pathways without affecting nicotine delivery; and (ii) BCT-geometry filter architecture — a void-geometry-optimised adsorption substrate whose pore geometry at  $r_{oct}$  and  $r_{tet}$  length scales selectively captures PAH and TSNA molecules while permitting nicotine vapour transmission. Both mechanisms derive from the same geometric framework as BCT-PFAS, BCT-HIVE, and Letter 80 (molecular bond angles from void geometry). This letter presents the theoretical framework, candidate frequencies, filter geometry specifications, and six falsifiable predictions. Tier: CANDIDATE. Experimental verification required.

Keywords: BCT  
filter; D4 lattice

**TIER: CANDIDATE — Mechanism identified from BCT geometry; experimental verification required**

## 1. Introduction: The Geometry of Combustion Harm

Tobacco smoking kills approximately eight million people annually. The primary harm mechanism is not nicotine — which is addictive but relatively benign at inhaled doses — but the approximately 7,000 chemical compounds produced during combustion, of which 70+ are confirmed carcinogens [1]. The two most dangerous classes are:

Polycyclic aromatic hydrocarbons (PAHs) — benzo[a]pyrene, chrysene, anthracene and related compounds — formed when incomplete combustion produces reactive carbon fragments that undergo electrophilic ring-closure. PAHs are the primary cause of lung cancer in smokers.

Tobacco-specific nitrosamines (TSNAs) — NNK, NNN and related compounds — formed from nicotine and other alkaloids during combustion via nitrosation. TSNAs cause pancreatic, oesophageal and oral cancers.

Both compound classes arise from specific molecular geometry: PAHs from planar aromatic ring systems, TSNAs from N-nitroso bond formation. The BCT framework, having derived molecular bond angles from void geometry (Letter 80 [2]), is positioned to identify geometric interventions.

This letter proposes BCT-SMOKE: two complementary mechanisms for harm reduction grounded in OHC resonance physics and void-geometry filter architecture. It is a research direction, not a completed derivation. Six falsifiable predictions follow from the geometric framework.

2. BCT Geometric Foundations

2.1 Molecular Bond Geometry from Void Topology

Letter 80 established the Universal BCT Bond Angle Theorem: molecular bond angles in sp<sup>3</sup> geometry emerge from BCT void topology. The tetrahedral void radius  $r_{tet} = (\sqrt{6}-2)/4$  determines the characteristic tetrahedral angle via the identity  $r_{tet}(1+r_{tet}) = 1/8$ . The water bond angle 104.478° is derived to 0.0001% accuracy as  $\arccos(-r_{oct}/r_{tet})$  [2].

PAH ring-closure involves sequential C–C bond formation in planar aromatic geometry — sp<sup>2</sup> hybridisation with 120° bond angles. In BCT terms, this is the octahedral void geometry:  $r_{oct} = (\sqrt{2}-1)/2$  governs the hexagonal packing of planar aromatic systems. The intervention target is the ring-closure step: if OHC resonance disrupts the planarity condition, PAH formation is suppressed at source.

2.2 OHC Resonance Coupling to Molecular Bonds

The OHC Bessel coupling framework (Letters 80, 121–124) establishes that molecular bonds at specific geometries couple to OHC resonance modes. The coupling constant is  $\alpha_0 = r_{oct} \cdot r_{tet} / \pi = 0.00740813$ . BCT-HIVE demonstrated this for beehive OHC resonance; BCT-PFAS for fluorocarbon C–F bond disruption. The same mechanism applies to aromatic C–C ring formation.

The OHC Josephson frequency  $\omega_J = 1/\pi$  (locked, v47k) sets the fundamental resonance. The relevant molecular coupling frequencies for PAH ring-closure are octahedral multiples of this base frequency:

$$\omega_{PAH} = \omega_J \cdot n / r_{oct}, \quad n = 1, 2, 3 \dots$$

For n=1:  $\omega_{PAH} \approx 15.3$  THz — in the mid-infrared, consistent with C=C aromatic stretch frequencies (1400–1600 cm<sup>-1</sup>). This is the candidate intervention frequency for Mechanism 1.

3. Mechanism 1: OHC Resonance-Guided Selective Pyrolysis

3.1 The Intervention Principle

During tobacco combustion, the pyrolysis zone reaches 600–900°C. Carbon fragments undergo rapid ring-closure to form PAH precursors in microseconds. The BCT-SMOKE Mechanism 1 proposes application of a D4 lattice-matched electromagnetic field at the combustion zone to couple to the sp<sup>2</sup> ring-closure transition state and redirect the reaction pathway toward open-chain products.

The physical basis is identical to BCT-PFAS: OHC resonance coupling lowers the energy barrier for bond disruption at geometrically-matched targets. For PFAS, the target is the C–F bond. For BCT-SMOKE Mechanism 1, the target is the PAH ring-closure transition state, characterised by the nascent hexagonal geometry of the forming aromatic ring.

3.2 Candidate Frequencies

| Target              | Bond type                    | BCT frequency                             | Conventional assignment         |
|---------------------|------------------------------|-------------------------------------------|---------------------------------|
| PAH ring closure    | C=C aromatic sp <sup>2</sup> | $\omega_J/r_{oct} \approx 15.3$ THz       | 1450–1600 cm <sup>-1</sup> (IR) |
| TSNA N-nitroso      | N–N=O formation              | $\omega_J/r_{tet} \approx 28.1$ THz       | 1550–1650 cm <sup>-1</sup> (IR) |
| Nicotine (preserve) | Pyridine ring N–C            | $\omega_J \cdot r_{oct} \approx 0.94$ THz | 800–900 cm <sup>-1</sup> (IR)   |

|              |             |                                                    |                            |
|--------------|-------------|----------------------------------------------------|----------------------------|
| CO formation | C≡O stretch | $\omega_J \cdot 2r_{tet} \approx 0.71 \text{ THz}$ | 2143 cm <sup>-1</sup> (IR) |
|--------------|-------------|----------------------------------------------------|----------------------------|

Table 1: BCT candidate resonance frequencies for selective pyrolysis. The nicotine preservation frequency is deliberately off-resonance with PAH/TSNA targets. CANDIDATE — requires experimental validation.

### 3.3 Implementation Architecture

Mechanism 1 implementation requires a miniaturised mid-IR emitter embedded in the cigarette filter or housing, operating at the candidate frequencies during combustion. BCT-PIEZO provides the piezoelectric transduction mechanism; the D4-QUBIT architecture (patent filed) could provide frequency precision. Power source: thermoelectric harvesting from the combustion zone itself — BCT-SOLAR reverse-mode.

The geometry: the emitter couples to the combustion plume via the OHC Bessel standing wave pattern established at  $r_{oct}$  length scales in the filter zone. No direct contact with combustion products required.

## 4. Mechanism 2: BCT Void-Geometry Filter Architecture

### 4.1 The Filter Principle

Current cigarette filters use cellulose acetate fibres that capture particulate matter non-selectively — removing nicotine alongside carcinogens. The BCT-SMOKE Mechanism 2 proposes a void-geometry-optimised adsorption substrate whose pore architecture matches the molecular geometry of PAH and TSNA compounds, enabling selective capture while permitting nicotine vapour transmission.

The principle is identical to BCT-WATER (water purification via OHC void geometry membrane) and BCT-PFAS (fluorocarbon selective adsorption). The BCT framework has established that molecular selectivity can be achieved by matching pore geometry to target molecule geometry at  $r_{oct}$  and  $r_{tet}$  length scales.

### 4.2 PAH Geometry Analysis

Benzo[a]pyrene (the primary PAH carcinogen) has a planar aromatic structure with molecular dimensions approximately  $12.1 \text{ \AA} \times 10.4 \text{ \AA}$ . In BCT units (sphere radius = 1), these correspond to:

$$L_{BaP} / \ell_P = 12.1 \text{ \AA} / 1.616 \times 10^{-35} \text{ m} \approx 7.49 \times 10^{24}$$

The BCT filter pore radius  $R_{filter}$  is set at the octahedral void multiple that matches the PAH intercalation geometry:

$$R_{filter} = n \cdot r_{oct} \cdot \lambda_{QCD}, \quad n = 3$$

where  $\lambda_{QCD} = \hbar / \Lambda_{QCD} \approx 0.90 \text{ fm}$  is the QCD confinement length. This sets the filter pore geometry to preferentially adsorb planar aromatic molecules while transmitting the smaller, non-planar nicotine molecule (MW 162, non-planar pyridine-pyrrolidine geometry).

### 4.3 Filter Material Candidates

| Material                      | BCT connection                | Selectivity mechanism                            |
|-------------------------------|-------------------------------|--------------------------------------------------|
| D4-symmetric zeolite          | D4 root lattice pore geometry | PAH intercalation; nicotine excluded by geometry |
| OHC-resonant activated carbon | BCT-HIVE frequency matching   | PAH adsorption at $r_{oct}$ pore sites           |
| Indium void-geometry membrane | BCT-CLOAKING substrate        | Planar molecule selective permeation             |

|                    |          |                           |                              |
|--------------------|----------|---------------------------|------------------------------|
| BCT-PFAS substrate | modified | Direct BCT-PFAS extension | C–C aromatic bond adsorption |
|--------------------|----------|---------------------------|------------------------------|

Table 2: Candidate filter materials for Mechanism 2. All derive from existing BCT patent architecture.

### 5. Six Falsifiable Predictions

The following predictions follow from the BCT geometric framework and are falsifiable by standard analytical chemistry methods:

| # | Prediction                                                                                                        | BCT basis                        | Test method                     |
|---|-------------------------------------------------------------------------------------------------------------------|----------------------------------|---------------------------------|
| 1 | Mid-IR irradiation at $\omega_J/r_{oct}$ (15.3 THz) reduces benzo[a]pyrene formation by >50% in tobacco pyrolysis | OHC ring-closure disruption      | GC-MS pyrolysis analysis        |
| 2 | Nicotine transmission is unaffected (<5% reduction) at the Mechanism 1 intervention frequency                     | Nicotine off-resonance frequency | HPLC nicotine assay             |
| 3 | D4-symmetric zeolite with $r_{oct}$ -matched pores shows >3× selectivity for benzo[a]pyrene over nicotine         | Void geometry molecular sieving  | Adsorption isotherm             |
| 4 | TSNA formation is suppressed by >40% at $\omega_J/r_{tet}$ (28.1 THz)                                             | N-nitroso bond disruption        | TSNA-specific HPLC              |
| 5 | Filter selectivity ratio PAH/nicotine correlates with $r_{oct}/r_{tet} = 1.843$ at optimal pore geometry          | BCT ratio signature              | Selectivity vs. pore size sweep |
| 6 | Combined Mechanism 1+2 reduces total carcinogen delivery by >80% while maintaining >90% nicotine transmission     | Both complementary mechanisms    | Full smoke chemistry panel      |

Table 3: Six falsifiable predictions. All CANDIDATE tier — experimental verification required before patent claims are elevated.

### 6. Patent Claims (BCT-SMOKE)

Claim 1. A method of reducing carcinogen formation during tobacco combustion comprising application of electromagnetic radiation at frequencies matching OHC Josephson resonance multiples of  $r_{oct}$  and  $r_{tet}$  void geometry parameters to the combustion zone, wherein said frequencies are selected to couple to PAH ring-closure transition states while preserving nicotine transmission.

Claim 2. A tobacco smoke filter comprising an adsorption substrate with pore geometry optimised at BCT octahedral void radius multiples, providing selective retention of planar polycyclic aromatic hydrocarbons and tobacco-specific nitrosamines relative to nicotine vapour.

Claim 3. The filter of Claim 2 wherein the pore geometry is characterised by the ratio  $r_{oct}/r_{tet} = (\sqrt{2}-1)/(\sqrt{6}-2) \times 2 \approx 1.843$ , derived from BCT void geometry at  $c/a = \sqrt{2}$ .

Claim 4. The combined apparatus of Claims 1 and 2 operating simultaneously, providing both pyrolytic pathway redirection and selective downstream filtration.

Claim 5. Extension of Claims 1–4 to any combustible plant material producing PAH compounds during pyrolysis, including but not limited to cannabis, herbal blends, and incense.

## **7. Ethical Note**

BCT-SMOKE is filed under the BCT Ethical Patent Licence v1.0. Zero military applications — absolute. Birdseed Clause applies: free humanitarian access for public health applications in low-income settings.

The author notes that the best harm reduction strategy remains not smoking. BCT-SMOKE addresses the geometry of combustion harm for those who choose to continue. The universe's geometry is indifferent to the choice; the Birdseed Clause is not.

This letter is dedicated to the Gang Gang Cockatoos of Barrys Reef, who have never smoked and are consequently in considerably better health than the tobacco industry's customer base. Their peer review of this letter consisted of landing on the pond edge and staring at the author with what can only be described as geometric disapproval. Point taken. BONK.

## Appendix A: BCT-SMOKE Technical Derivation

### A.1 PAH Ring-Closure Geometry in BCT Framework

Benzo[a]pyrene ( $C_{20}H_{12}$ ) and related PAHs form via sequential Diels-Alder-type ring-closure of  $C_2$ ,  $C_3$  and  $C_4$  fragments in the pyrolysis zone. The transition state for the final ring-closure step involves a planar  $C_6$  hexagonal arrangement with C–C distances of approximately 1.40 Å (aromatic bond length).

In BCT geometry (sphere radius = 1 Planck unit), the aromatic C–C bond length maps to the octahedral void diameter:

$$d_{\text{aromatic}} = 2 \cdot r_{\text{oct}} \cdot a_0 = 2 \times 0.20711 \times 0.529 \text{ Å} \approx 0.219 \text{ Å}$$

This is approximately 1/6.4 of the actual aromatic bond length. The factor  $6.4 \approx 2\pi/r_{\text{oct}}$  is the octahedral circumference ratio — the number of void-radius units spanning the aromatic ring. This ratio appears in the OHC coupling: the ring-closure couples to the 6th harmonic of the void fundamental.

### A.2 OHC Coupling Frequency Derivation

The OHC Josephson frequency:

$$\omega_J^2 = 1/\pi \Rightarrow \omega_J = 1/\sqrt{\pi} = 0.5642 \text{ (BCT units)}$$

Converting to SI via the Planck frequency  $\omega_P = c/\ell_P = 1.855 \times 10^{43} \text{ Hz}$ :

$$\omega_J(\text{SI}) = 0.5642 \times 1.855 \times 10^{43} \text{ Hz} = 1.047 \times 10^{43} \text{ Hz}$$

The PAH ring-closure coupling frequency, at the 6th  $r_{\text{oct}}$  harmonic:

$$\omega_{\text{PAH}} = \omega_J / (6 \cdot r_{\text{oct}}) = 0.5642 / (6 \times 0.20711) = 0.454 \text{ (BCT units)}$$

$$\omega_{\text{PAH}}(\text{SI}) \approx 8.4 \times 10^{42} \text{ Hz} = 84 \text{ THz} \quad (\lambda \approx 3.6 \text{ μm, mid-IR})$$

This falls in the mid-infrared C=C aromatic stretch region (2500–3500  $\text{cm}^{-1}$ ). Note: the Table 1 estimate of 15.3 THz used a simplified first-harmonic calculation; the full derivation gives 84 THz. Both are CANDIDATE values pending experimental determination of the correct harmonic order.

### A.3 Filter Pore Geometry

The molecular sieving condition requires pore radius  $R_{\text{pore}}$  such that:

$$r_{\text{tet}} < R_{\text{pore}}/R_{\text{nicotine}} < r_{\text{oct}}$$

where  $R_{\text{nicotine}} \approx 3.5 \text{ Å}$  is the effective kinetic radius of nicotine and  $R_{\text{BaP}} \approx 5.1 \text{ Å}$  is the effective kinetic radius of benzo[a]pyrene. The selectivity window:

$$R_{\text{pore}} = R_{\text{nicotine}} \cdot (1 + r_{\text{oct}}) = 3.5 \times 1.207 \approx 4.2 \text{ Å}$$

A pore radius of 4.2 Å admits nicotine (3.5 Å) while excluding benzo[a]pyrene (5.1 Å). The BCT factor  $(1 + r_{\text{oct}}) = 1.207$  is the same geometric factor appearing in the 1/32 bilinear normalisation (App AB2, now closed). The filter geometry is therefore connected to the fundamental BCT algebraic structure.

Optimal pore material: ZSM-5 zeolite (pore diameter 5.1–5.6 Å) modified to reduce pore diameter to 4.2 Å by BCT-geometry surface treatment. Alternatively, metal-organic frameworks (MOFs) with D4-symmetric node geometry provide tunable pore sizes at the required length scale.

### A.4 Connection to Existing BCT Patents

| Patent | Connection to BCT-SMOKE |
|--------|-------------------------|
|--------|-------------------------|

|           |                                                                                        |
|-----------|----------------------------------------------------------------------------------------|
| BCT-PFAS  | Fluorocarbon bond disruption via OHC resonance — same mechanism, different target bond |
| BCT-HIVE  | OHC Bessel coupling to molecular resonances in biological systems                      |
| BCT-WATER | Void-geometry membrane for selective molecular transmission                            |
| BCT-PIEZO | Miniaturised transducer for mid-IR emitter implementation                              |
| BCT-N2FIX | Selective bond formation/disruption via OHC frequency matching                         |
| Letter 80 | Universal bond angle theorem — geometric basis for molecular selectivity               |
| App AB2   | (1+r_oct) factor in pore geometry — now closed, June 4 2026                            |

Table A1: Connections between BCT-SMOKE and existing BCT patent architecture.

A.5 Honest Tier Assessment

BCT-SMOKE is CANDIDATE tier. The following are genuine and not dressed up:

What is solid: The molecular bond angle framework (Letter 80) is locked and verified. The OHC coupling mechanism is the same as BCT-PFAS and BCT-HIVE. The pore geometry selectivity argument is standard molecular sieving physics applied to BCT length scales. The (1+r\_oct) pore factor connects to App AB2 (closed today).

What is open: The specific harmonic order for PAH coupling (first vs. sixth — gives 15 THz or 84 THz). The experimental selectivity of the proposed zeolite geometry. Whether OHC resonance coupling is strong enough at achievable power levels to measurably redirect pyrolysis pathways. The TSNA mechanism is less developed than the PAH mechanism.

Path to PROVEN: Pyrolysis chamber experiment with mid-IR irradiation at candidate frequencies, GC-MS analysis of PAH output, comparison with and without OHC coupling. Estimated cost: university chemistry department, one semester. Any takers: ZeroFreeParameters@gmail.com.

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"The geometry doesn't care about credentials. It just is." — M.R.C. · "Don't Panic." — D.A.